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A freight transport optimization model for integrated network, service, and policy design

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ABSTRACT

This paper presents a freight transport optimization model that simultaneously incorporates multimodal infrastructure, hub-based service network structures, and the various design objectives of multiple actors. The model has been calibrated and validated using real-life data from the case study of hinterland container transport of the Netherlands, where CO_2 pricing, terminal network configuration, and hub-service networks are chosen as the design measures. Policy packages combining multiple types of policies show better network performance as compared with the optimal performance resulting from a single policy type. This illustrates the value of incorporating multiple types of policies simultaneously in freight transport optimization.

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1. Introduction

Shifting freight flows from road to more sustainable transport modes such as rail, inland waterway and sea transport, has been recognized by the European Commission (EC) as an important policy strategy "to establish a sustainable transport system that meets society's economic, social and environmental needs" (CEC, 2006, 2009; EC, 2001). For decades, there has been continuous effort to promote and implement such a modal shift (CEC, 2006). However, the available statistics indicate that the volumes shifted from road to the other above-mentioned transport modes have been limited with the expectation of the modal split stabilizing in the longer term (CEC, 2006). Under such circumstances, the EC Freight Transport and Logistics Action Plan (CEC, 2007) emphasizes that more efforts are needed to achieve a more substantial modal shift. The latest policy strategy of the White Paper on Transport of the European Commission (EC, 2011) includes infrastructure development, service quality enhancement, and regulatory measures.

In order to evaluate the combined effects of various policy strategies applied to different regions or different actors, design and evaluation approaches need to be developed that simultaneously incorporate infrastructure, services, and regulation at a large spatial scope.

Freight transport system has been studied extensively (Caris et al., 2013; Crainic, 2000). However, one of the main contributions of this paper has been lacking: inclusion of the governmental perspective by considering the social, economic, and environmental issues that inherently increase network design complexity, and bring new challenges for solutions from the following perspectives.

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Firstly, in strategic network design from a governmental perspective, the spatial scope of policy impacts is often enlarged to the national or international level. Consequently, multiple transport modes are available, thus requiring that the choice of mode and terminal be considered when solving routing problem(s). While evaluating different transport options, the potential cost efficiencies realized by better utilization of capacities of corridors, terminals, fleets, and vehicles need to be considered. Secondly, multiple types of commodities bring heterogeneous criteria in the choices of network use. Different commodities with different appearances and values could require transport networks of diverse modes, speed, costs, reliability, and security. Finally, the multiple actors participating in freight transport can introduce different design objectives into decision making, and introduce possibilities for cooperation or competition.

In addition to evaluating the impacts of different types of policies simultaneously, the model proposed in this paper contributes to dealing with the above-mentioned complexities and challenges of freight transport network design.

Substantial research has been carried out in developing appropriate methods for assessing the performance of freight transport policies. For example, Beuthe et al. (2002) simulated the impacts of internalizing the external costs on multimodal freight flows over a real-life network. Groothedde et al. (2005) developed a many-to-many hub-based service network design model for distributing fast-moving consumer goods. Yamada et al. (2009) introduced a multimodal freight transport network design model for minimizing the network costs from the governmental perspective, and the route costs from the network users' perspective. Limbourg and Jourquin (2009) developed a hub allocation model, which provided the optimal locations of the European intermodal terminals given a certain number of hubs and the candidate locations of these hubs. The necessity of integrating the service networks was also recognized through an extension of this model (Jourquin et al., 2009a,b). Yamada et al. (2011) presented a network design model by considering the transport network and the supply chain network simultaneously. A review of the multimodal freight transport models developed since 2005 has been compiled by SteadieSeifi et al. (2014). However, none of these models has been able to evaluate infrastructure, service network development, and regulatory policies simultaneously.

This paper aims to fill this gap by focusing on the interaction between the infrastructure network, service network, and regulatory policies. The model presented could be used for strategic network design in a medium- to long-term, mainly considered from a governmental perspective.

In support of this approach combining the three design dimensions, we have developed an extended bi-level optimization model that combines the demand and supply side of the freight transport system with two main innovative features. The novelty of the supply-side model is the interaction between infrastructure and the service network, while an important novelty on the demand side is the multimodal route choice model which enables terminal choice and has been calibrated using the transhipment data of intermodal terminals. We are not aware of any literature that reports on the empirical validity of a freight virtual network model for actual transhipment volumes. The application of the combined approach to a case study for the Dutch hinterland container network has resulted in original findings concerning the sensitivity of overall network performance to the three design dimensions.

In addition to this introductory section, the paper consists of three other sections. Section 2 presents the mathematical structure of the proposed model. Section 3 explains the case of hinterland container transport in the Netherlands, as well as the data preparation, model calibration and validation of this application. We presents six cases of application and a number of implications for the policy makers derived from the above-mentioned applications. Section 4 contains our conclusions and discussion of the results including potential directions for further research.

2. Model specification

2.1. Basic structure of the model

In terms of its basic structure, the proposed model can be considered as a multi-actor, multi-commodity, multimodality freight transport network optimization model. It captures the objectives of the particular actors/stakeholders involved such as the government authorities at different institutional levels, terminal, and transport operators. Government objectives are assumed to be reduction of the total network costs and the CO_2 (Carbon Dioxide) emissions through the terminal network configuration and CO_2 pricing. The terminal operators' aims are assumed to be attracting more freight flows, achieving economies of density and economies of scale, and thus providing services at lower costs. Transport operators are assumed to collaborate, and if possible, operate hub-based inland waterway transport services in order to save costs by improving the efficiency of capacity utilization. Shippers are assumed to choose the least expensive transport mode, terminal, and route in order to reduce their costs.

The model is formulated as a bi-level optimization model with two pre-processing procedures. The objective of government is reflected at the upper-level of the problem, which searches for the optimal alternatives leading to the optimal network performances represented by the minimum total network generalized costs and CO_2 emissions. The lower-level assigns the multi-commodity flow to the multimodal network, thus minimizing the users' costs based on the costs of each (possible) route.

The objectives of terminal and transport operators are taken into account through two pre-processing procedures that are linked to the flow assignment at the lower-level of the problem. One pre-processing procedure minimizes the service costs considering economic fleet size, and barge size, given the predicted demand for hub-based barge services, while the other

calculates the flow-related transhipment costs of the terminals while taking economic terminal capacity into account. These two pre-processing procedures and the flow assignment procedure are performed iteratively in order to achieve equilibrium in flow assignment at the network level. A simplified flow chart is shown in Fig. 1.

The network performances are optimized through scenario-based optimization. This implies that the model generates a large number of scenarios, evaluates their performances, and searches for the best. A scenario may consist of one or a combination of different design measures, namely infrastructure modifications, regulatory policies, and potential hub-based services. In order to simplify the formulation, a specific example to explain the logic and the mechanism of the mathematical specification of this model is taken. In this example, it is assumed that each scenario consists of a certain terminal network configuration with (or without) feasible hub-based barge services, and a CO_2 charge.

2.2. The upper-level problem

The network performances are evaluated using the total network costs defined as the total costs of moving all freight volumes from their origins to destinations along the lowest-cost route(s). The total network costs consist of the internal (i.e. operational) costs and charges of CO_2 emissions for all links in the network. The charges of CO_2 emissions are expressed in relative monetary terms €/tCO_2 (euro per ton CO_2 emissions). The optimization problem is set up as follows.

Minimize the objective function:

$$\text{MIN Total network costs} = f(\mu, C^*, f^*, c^{co_2}), \tag{1}$$

Subject to the following constraints:

$$\mu_{w,b} = \begin{cases} 0 \\ 1 \end{cases} \tag{2}$$

$$\sum C_w^* \leqslant \sum C_w \tag{3}$$

where

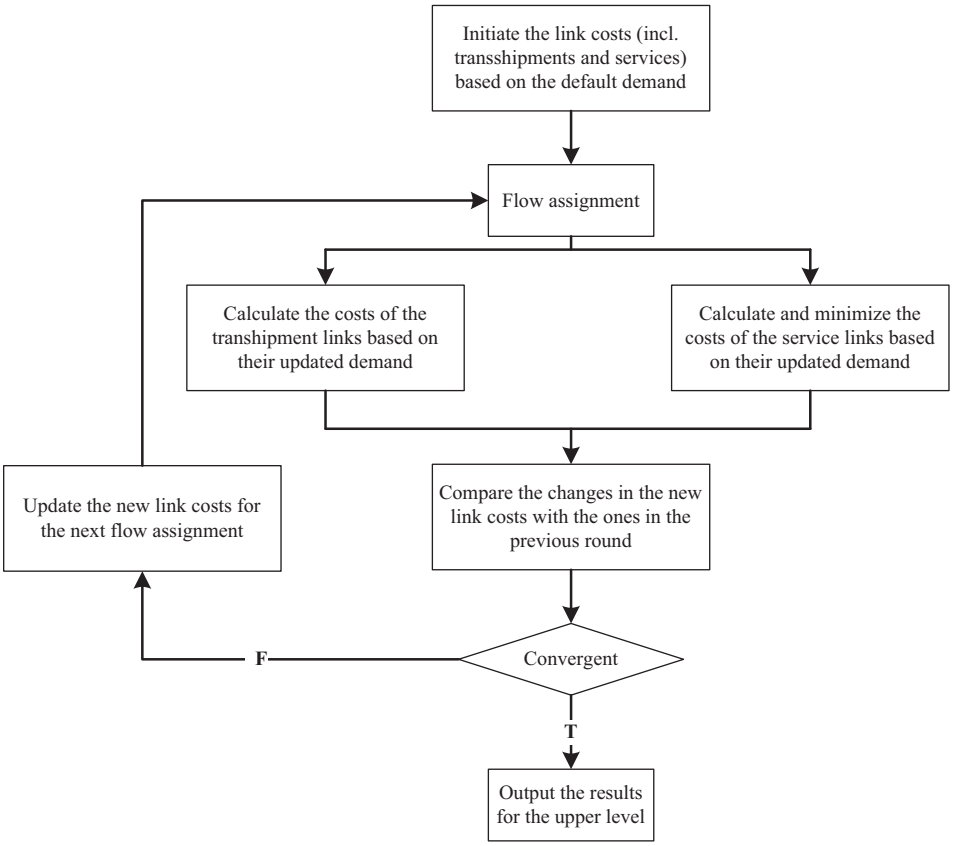


Fig. 1. Flow chart of the flow assignment procedure considering the flow-related transhipment costs and hub-based-service costs.

Sets	
B	is the set of candidate terminals with given locations, which are the elements used to compose an alternative terminal network configuration;
W	is the set of alternative terminal network configurations.
Variables of the model	
μ	is a binary array expressing the openness of terminals;
$\mu_{w,b}=1$	if terminal b is opened in terminal network configuration $w, b \in B$, where B is the set of candidate terminals, and $w \in W$, where W is the set of alternative terminal network configurations (2^n configurations);
C^*	is the vector of total link costs of the optimal solution at the lower-level;
f^*	is the vector of link flows of the optimal solution at the lower-level;
c^{co_2}	is the price of CO_2 emissions (€/tCO_2).

The upper-level minimizes the total network costs. Constraint (2) defines that a terminal can be either open (in operation) or closed (out of operation). Constraint (3) ensures that the "optimal solution" at the lower-level is a solution which is not worse than any other solution.

2.3. The lower-level problem

The multi-commodity flow is assigned to the multimodal network at the lower-level. Shippers are assumed to choose the mode, terminal, and route of their shipments. Given the fixed transport demand, it is assumed that they make choices based on the minimum route costs. An advancement of this model is that the mode, transport terminal, and route are chosen concurrently, which is facilitated by using the virtual network representation. The concept of virtual network has been applied to design of the freight transport network in earlier research (e.g. Jourquin, 1995). The novelty of the network representation in this model is that in addition to representation of the road, rail, IWW (Inland Waterway) links, access/egress, pre-/end-haulage, and/or the transhipment links are also taken into account. The virtual network is further extended by including the hub-based barge service links, which represent collaborative hub-based barge services. Thus hub-based barge services are differentiated from the shuttle barge services. The new virtual network facilitates the design of both the infrastructure and service network while simultaneously taking into account economies of scale and economies of density of both terminals and hub-based services. The detailed mathematical/analytical representation of the network is given in Appendix A.

The costs of using each type of links are defined based on real-life cost structures.

- Transport mode-related costs, commodity-related costs, and CO_2 emission charges are components of the costs of road, rail and/or shuttle barge service links. Mode-related costs are further specified in two components: (i) the distance-related component, which depends on the distance between the origin and the destination, including maintenance, repairs, and fuel; and (ii) the time-related component representing the time costs directly imposed on the transport operators of different modes including depreciation, interest, facility tax, licences and permits, insurance, wages, and business costs.
- Commodity-related time costs mainly including interest, depreciation, and reduction in the market value of the freight during transportation independently of the transport mode. (Since shippers dealing with high commodity-related time costs are more likely to choose a fast transport mode, commodity-related time costs are used to expose the heterogeneity of multi-commodity for choice of mode.)
- Costs of terminal operators including handling costs depending on the terminal scale and throughput reflecting economies of scale and density (Ballis and Golias, 2002). (A monotonic decreasing function has been used to handle variations in the handling costs.)
- Costs of access/egress links comprise the average transport costs of truck haulage between any location within the region and the highway entrance.
- Costs of pre-/end-haulage relate to the costs of truck haulage connecting a location within a region to a terminal. (Error terms are added to each transhipment, access/egress, or pre-/end-haulage link to compensate for the phenomenon where the costs of different terminals or regions may be different due to local conditions or other causes of variations that are difficult to measure.)
- The costs of hub-based barge service links depend on the volume of transport demand, barge size, service frequency, and other factors. Additionally, these costs are highly dependent on the structure of service network as shown by the examples of the structures given in Appendix B. Due to the inherent complexity of the link cost function in the given context, a generic analytical cost function showing the relevant variables and parameters is presented as follows.

The total costs of link l as a leg of the service S_i are:

$$C_s^{\lambda,p} = f\left(c_s^{\lambda,F}, c_s^{\lambda,D}, c_s^{\lambda,U}, nr_s^{\lambda}, z_s^{\lambda}, k_s^{\lambda}, v_s^{\lambda}, u_s^{\lambda,H}, u_s^{\lambda,SH}, f_s^{\lambda}, d_s^{\lambda}, \text{sns}\right), \quad \forall s \in S_i, S_i \subset \text{SN} \tag{4}$$

where

Sets

SN is the set of links contained in the hub-based service network;
S_i is the set of links contained in the hub-based-service $S_i \subset SN$.

Parameters

$c_s^{\lambda,F}$ is the annual fixed costs of a barge operating along link s, $c_s^{\lambda,F} = f(z_s^{\lambda})$;
$c_s^{\lambda,D}$ is the distance-related variable costs of moving freight along link s, (€/t-km), $c_s^{\lambda,D} = f(z_s^{\lambda}, v_s^{\lambda})$;
$c_s^{\lambda,U}$ is the time-related variable costs of moving freight along link s (€/t-h), including mode-related time costs and commodity-related time costs, $c_s^{\lambda,U} = f(z_s^{\lambda})$;
$z_s^{\lambda*}$ is the optimal barge size along link s considering navigability of the link and utilization of barge capacity;
v_s^{λ} is the average speed of barges operating along link s;
$u_s^{\lambda,H}$ is the total handling time of a round trip along link s;
$u_s^{\lambda,SH}$ is the total shipping time of a round trip along link s;
d_s^{λ} is the length of link s;
sns is the service network structure;

Variables

$nr_s^{\lambda*}$ is the optimal number of barges needed to serve a given volume of demand over link s, $nr_s^{\lambda} = f\left(z_s^{\lambda*}, f_s^{\lambda}\right)$;
f_s^{λ} is the freight flow (ton) along link l_i;
k_s^{λ} is the total annual transport capacity of the fleet of $nr_s^{\lambda*}$ barges (full load and the maximum number of turnarounds, $k_s^{\lambda} = f\left(z_s^{\lambda*}, v_s^{\lambda}, d_s^{\lambda}\right)$;

In Eq. (4), the volumes of transport demand and the required service level represented by the service frequency determine the unit transport costs of a service. The cost function has been simplified by introducing the following assumptions.

- All barges operating on a service link have the same loading capacity (measured in TEUs).
- The barges deployed are maximally utilized at the annual scale in terms of the number of turnarounds.
- The annual fixed costs of operating a barge $(c_s^{\lambda,F})$, the distance-related variable costs $(c_s^{\lambda,D})$, and the time-related variable costs $(c_s^{\lambda,U})$ are independent from the barge's load factor.
- The annual fixed costs include the costs of depreciation, maintenance, and interest charged on the capital.
- The distance-related variable costs mainly include fuel costs.
- The time-related variable costs mainly refer to the crew costs.

Given the definition of the cost function of each type of links, the lower-level optimization problem is set up as follows. Minimize the objective function:

$$\text{MIN} \quad \sum_{i,j,o_1,o_2,p} f_{i,j}^{o_1,o_2,p} \cdot C_{i,j}^{o_1,o_2,p} \cdot \delta_{i,j}^{o_1,o_2,p}, \quad \forall (i,j) \in L, o_1 \in O, o_2 \in O, p \in P \tag{5}$$

Subject to the following constraints:

$$C_{i,j}^{o_1,o_2,p} = f^1\left(f_{i,j}^{o_1,o_2,p}\right)\Big|_{(i,j)\in H}, \quad \forall (i,j) \in H; \tag{6}$$

$$C_{i,j}^{o_1,o_2,p} = f^2(f_{i,j}^{o_1,o_2,p})\Big|_{(i,j)\in S}, \quad \forall (i,j) \in S; \tag{7}$$

$$\delta_{i,j} = 0|_{\delta_i=0}, \quad \forall (i,j) \in L, \ i \in T; \tag{8}$$

$$\delta_{i,j} = \delta_{j,i}, \quad \forall (i,j) \in L, \ (i,j) \in L; \tag{9}$$

$$\delta_{i,j}^{o_1,o_2,p} + \delta_{j,l}^{o_1,o_2,p} \leqslant 1|_{(i,j)\in H \text{ and } (j,l)\in H}, \quad \forall (i,j) \in L, (i,l) \in L; \tag{10}$$

$$\delta_{i,j}^{o_1,o_2,p} + \delta_{j,l}^{o_1,o_2,p} \leqslant 1|_{(i,j)\in G \text{ and } (j,l)\in G}, \quad \forall (i,j) \in L, (j,l) \in L; \tag{11}$$

$$\delta_{i,j}^{o_1,o_2,p} + \delta_{j,l}^{o_1,o_2,p} \leqslant 1|_{(i,j)\in A \text{ and } (j,l)\in A}, \quad \forall (i,j) \in L, (j,l) \in L; \tag{12}$$

$$\sum_i f_{i,o_1}^{o_1,p} = f^{o_1,p}, \quad \forall (i,o_1) \in A, o_1 \in O, \ p \in P; \tag{13}$$

$$\sum_i f_{i,o_2}^{o_2,p} = f^{o_2,p}, \quad \forall (i,o_2) \in A, o_2 \in O, \ p \in P; \tag{14}$$

$$\sum_{i,j} f_{i,j}^{o_1,p} = \sum_{i,j} f_{i,j}^{o_2,p}, \quad \forall (i,j) \in L, o_1 \in O, \ o_2 \in O, \ p \in P; \tag{15}$$

$$f_{i,j}^{o_1,o_2,p} \geqslant 0, \quad \forall (i,j) \in L; \tag{16}$$

where

Sets

L	is the set of links;
H	is the set of links representing transhipment between terminal and rail/inland waterway, $H \subset L$;
A	is the set of links representing access/egress, $A \subset L$;
G	is the set of links representing pre- and end-haulage, $G \subset L$;
S	is the set of links representing links of hub services, $S \subset L$.
O	is the set of regions; and
T	is the set of terminals;

Variables

$f_{i,j}^{o_1,o_2,p}$	is the flow of commodity p from origin o_1 to destination o_2 over link (i,j);
$C_{i,j}^{o_1,o_2,p}$	is the total costs of link (i,j) when moving one unit of commodity p from origin o_1 to destination o_2; and
$\delta_{i,j}^{o_1,o_2,p}$	is the availability of link (i,j) for moving a unit of commodity p from origin o_1 to destination o_2; $\delta_{i,j}^{o_1,o_2,p} = 0$ when the link is not available, $\delta_{i,j}^{o_1,o_2,p} = 1$ otherwise.

The total network costs are minimized by assigning the transport demand over the network with a given terminal configuration and CO_2 charge. Eqs. (6) and (7) indicate the dependency of the transhipment costs and that of the service link(s) on the flows they handle. The total flows they handle are treated as parameters in this combined route choice model, which is independent from the flow, denoted as $f_{i,j}^{o_1,o_2,p}$, of commodity p from origin o_1 to destination o_2 over link (i,j). Constraint (8) prohibits assigning flows to 'closed' terminals, while constraint (9) defines the availability of links in both directions. To prevent unrealistic shortcuts, the flows are not allowed to be moved over two adjacent access/egress links (Constraint (12)). The same rule applies to the pre-/end-haulage and transhipment links between terminals and rail/inland waterways (Constraints (10) and (11)). Constraints (13) and (14) ensure that all flows are assigned over the network. Constraint (15) balances the flow generation and attraction at the network level, while constraint (16) prevents any negative flows over the network. Bearing in mind that the model is developed for long-term strategic planning, it is assumed that investments are made when needed. Consequently, the capacities of fleets, transport infrastructure, and terminals can be extended continuously to meet the increased demand. Therefore, the capacity extension has no impact on the assumed cost functions.

2.4. Solution methods and algorithms

A derivative all-or-nothing (AON) algorithm based on the Dijkstra method (Dijkstra, 1959) is used for flow assignment according to the lowest total route costs. An AON is chosen due to the fact that the capacities of the candidate routes are assumed to be unconstrained. This assumption is made since annual aggregated flow data are often used in strategic networks. At this aggregated level, capacity shortages are rarely observed. Since congestion is mostly observed in urban areas, it is captured by the higher distance-related costs of the access/egress links and pre-/end-haulage links in this model, and the differences among different urban areas are partially captured by alternative specific constants associated with these links. Using the AON algorithm saves a lot of computing time as it eliminates the need for redundant iterative procedures for achieving the equilibrium.

At the upper-level, the combinatorial optimization problem is solved using a genetic algorithm (GA) to find the combinations of design measures that result in the minimal total network cost (Goldberg, 1989). In this case the GA is designed by applying the 'roulette wheel selection', 'n-point half-uniform crossover', 'uniform mutation', and 'elitist' strategies (Christopher et al., 1995; Costa and Oliveira, 2004; Spears and De Jong, 1991; Wardlaw and Sharif, 1999). The optimization steps are given as follows.

Inner loop: flow assignment

Step 1: Initiate the network features according to the reference scenario.

Step 2: Load solution $j, j \in [1,$ *number of solutions of this generation*$]$ of generation i of the network. The transhipment costs of each terminal and the transport costs of each service leg are given according to the costs in the base year.

Step 3: Run the multi-commodity AON flow assignment.

Step 4: Update the transhipment costs of each terminal based on the transhipment cost-flow function and the flow assigned to the terminal in this run; meanwhile, calculate the costs of each service leg, as presented in Eq. (4) of the service simulation sub-model.

Step 5: Iterate Step 3 and Step 4 until $\frac{\sqrt{\left(\sum_{l=0}^{n}(f^{t+1}-f^{t})^2/n\right)}}{\left(\sum_{l=0}^{n}f^{t}\right)/n} < 5\%,$ or $t = 20$,

where f is the total flow over link l, $(l \in H)$; n is the total number of candidate terminals; t is the number of iteration, $t \in [1, 20]$.

Step 6: Calculate the system costs and the CO_2 emissions of this solution.

Step 7: If $j + 1 =$ *the number of solutions of this generation*, then go to Step 2 of the outer loop.

Step 8: Load solution $j+1$ of generation i to the network, and then go to Step 3.

Outer loop: optimization

Step 1: Generate the first generation of solutions. Go to the inner loop for solution 1 of this generation.
Step 2: Find the "best" solution of this generation.
Step 3: If $i+1$ = *the number of generation*, then go to Step 6.
Step 4: Generate the next generation.
Step 5: Iterate Step 2 to Step 5.
Step 6: Return the "best" of all solutions of all generations, and stop.

3. An application of the model

3.1. The case – hinterland container transport in the Netherlands

The Port of Rotterdam (The Netherlands) has the largest annual container throughput in Europe. More than 7 million TEUs (TEU – Twenty-Foot Equivalent Unit) hinterland containers were handled by the port in 2010, with a modal split of 47%, 15%, and 33% by road, rail, and inland waterways, respectively (Port of Rotterdam, 2013). Based on the growth of maritime container transport and the ongoing capacity expansion of the port, the total demand for container transport is estimated to increase by 150% by 2020 (De Langen et al., 2012) compared to that in 2010. While contributing to the Dutch economy, handling such container traffic has also caused road congestion and related environmental impacts. Consequently, due to the increasing severity of the problem, the Dutch government has made substantial effort to develop and promote intermodal transport as a counterpart to the uni-modal and particularly road transport mode. A component of these efforts has been providing subsidies to the operating and expanding public intermodal terminals, and also for starting up new ones (Dutch Ministry of Infrastructure and the Environment, 2002). Internalizing the external costs by charging CO_2 emissions is another policy measure still under discussion by the government and public. In line with the anticipated increase in demand for transporting maritime containers (De Langen et al., 2012), sea terminal operators are developing inland/barge service networks in order to extend gate services and enhance their competitiveness in the future hinterland distribution (Notteboom, 2009; Port of Rotterdam, 2013). Hinterland transport operators have initiated plans to co-operate hub-based in addition to the current point-to-point (shuttle) barge transport services, and share their fleets. The aim is to increase competitiveness in the market by increasing service frequency and load factor (Brabant Intermodal, 2013; Visser et al., 2012). The above-mentioned measures cover terminal network configuration, service network design, and CO_2 pricing.

The new model is applied to evaluate the effects of each design measure and their combined effects. In such context, service-based measures are applied only where cooperation between transport operators is possible. The impact of each measure and their combinations is evaluated from the perspective of reducing emissions of CO_2 and the total (internal and external) network costs.

3.2. Inputs

The model is implemented with a geographic focus on the Netherlands and its connections to other European countries where containers are transported to/from/passing through the Netherlands via road, rail, or inland waterway. Real-life data are used in this application.

On the demand side, the transport demand across all Dutch NUTS3 regions (Eurostat, 2006) and other European regions is aggregated (the source is the freight transport survey provided by the Central Statistics Bureau of the Netherlands CBS (2006)). Since the case study focuses on container transport, the multi-commodity, as an important feature of freight transport, is not presented by the appearances of the freight (e.g. dry-bulk, liquid-bulk, break-bulk). Instead, the commodities are grouped into five groups according to the volume-to-value distribution of the containers transported to/from/through the Netherlands (see Fig. 2). This distribution is estimated from a sample derived from the international trade tables provided by Statistics Netherlands (StatLine, 2006).

On the supply side, the geographic information on the infrastructure networks was adapted from the GIS database of TransTools (EC JRC IPTS, 2005). The information for the Dutch terminals was obtained from Rail Cargo (Rail Cargo, 2011), and the Expertise and Innovation Centre Inland Waterway (EICB, 2013). The information for the other European inland terminals was collected from the websites of the terminals and other sources. The transport costs of each mode were estimated on the basis of previous research (Black et al., 2003; Decisio, 2002; NEA, 2003, 2004), and were adjusted in light of our enquiries made with particular service operators. The commodity-related time costs per commodity type were estimated by applying a regression function which relates the freight value in trade (Ponga, 2006) to the commodity-related time costs (De Jong et al., 2004; Kreutzberger, 2008). The handling costs of Dutch terminals are estimated at various scales based on regressing the actual handling prices and their throughput. The regression relationship shows strong scale effects in the typical range of operation – between 10,000 TEU/year and 500,000 TEU/year (see Fig. 3). Assuming that a similar regression applies to other European countries, the handling costs of non-Dutch container terminals can be estimated. Information

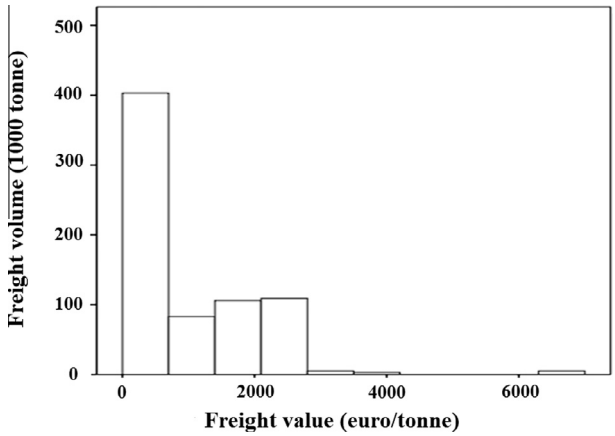


Fig. 2. Weight to value distribution of Dutch export freight. (A sample representing the statistic data of the freight exported from the Netherlands in 2006 calculated from the international trade tables provided by Statistics Netherlands (StatLine, 2006).)

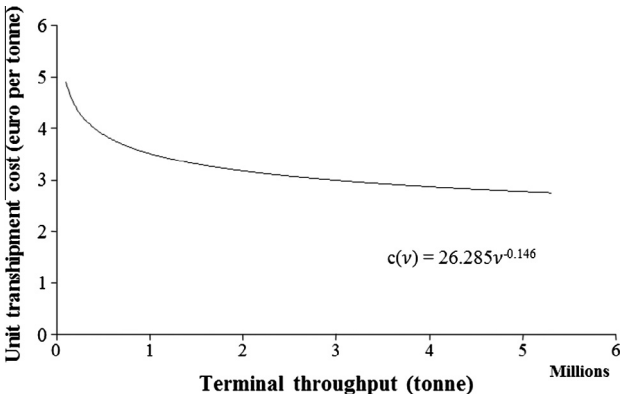


Fig. 3. Relationship between the average unit transhipment costs and the terminal scale.

on the network structures, fleet, and service frequency of Dutch hub-based barge services are collected from the Bureau of Inland Navigation Promotion and individual terminal operators.

Emissions of CO_2 from road, rail, inland waterway transport, and transhipment at terminals are calculated based on previous research (Geerlings and Van Duin, 2011; NEA et al., 2001). The emissions for different classes of barges are estimated for different speeds and drafts with a correction for shallow-water effects (Holtrop and Mennen, 1982).

3.3. Calibration and validation

Because the costs of each access/egress, pre-/end-haulage, and transhipment link depend on local conditions, unknown variations may occur in real-life. The error terms in the cost functions of these links are therefore calibrated. Model calibration is carried out by solving a parameter estimation problem, and searching for the optimal values for these parameters. These values, in combination, lead to a satisfactory solution. Based on the available observations, the results of calibration are evaluated at the regional, terminal, and link level. At the regional level, the results of the modal split per region (75 regions) are compared to those gained from the Dutch freight transport survey (CBS, 2006) (R^2 = 0.91). At the terminal level, the modelled throughput of terminals (28 terminals) is compared to the actual throughput of those terminals (R^2 = 0.88). At the link level, the results of relevant link flows are compared to the corresponding ship lock counts (27 ship locks) and the border counts (7 border crossing points) of the railway transport crossing the border of the Netherlands (R^2 = 0.81).

The model is validated by three additional steps. First, the flow over the network estimated by the calibrated model for the base scenario in 2006 represents the real situation. Second, most individual parameters are shown to stabilize within an acceptable range during multiple calibration runs. Last, the cross-elasticity of mode-related costs to the total network flows of each mode shows that the sensitivity of the model to the cost changes are acceptable including responses of the model to changes in costs (Yamada et al., 2009).

3.4. Results

Six analyses are carried out in order to evaluate the impacts of different network design measures, and their combinations in the given case. Table 1 gives the main characteristics of the policies evaluated in these (six) analyses. CO_2 pricing, terminal network configuration, and hub-service networks are chosen as the design measures to be analyzed for their relevance to the current practice of Dutch container transport. Each type of design measure is evaluated independently in the first three analyses. The combinations of two types of design measures are analyzed in Analysis 4 and 5. The combination of terminal network configuration and hub-service-networks are not evaluated due to the model's constraints. A hub-service-network is available only if all of the involved terminals are in operation, thus the results regarding the efficiency or effectiveness of this combination of design measures are not comparable to the other design measures. Analysis 6 evaluates all types of design measures simultaneously, with the constraint that a hub-service-network is available only if all of the involved terminals are in operation.

In the pool of potential design measures, which are used as inputs for the model, the CO_2 price is given in the range of 0–1000 at increments of 10 €/tCO_2 (101 scenarios). Although the high end of this range seems unfeasible from a policy point of view, including such a wide range is important in order to obtain an understanding of the system's responsiveness to variations in overall transport costs and modal shares. A total of 42 candidate terminal locations within the Netherlands have been defined on the basis of the existing hinterland container transport facilities of the Netherlands as the elements of the alternative terminal networks. A hub-service-network consisting of 15 terminals is defined with either of two statuses: in operation or not in operation. Due to the computation time, a given number of 30,000 or 50,000 scenarios are evaluated in an optimization process in the analyses where terminal network configurations are included.

3.4.1. Analysis 1: The CO_2 pricing policy

Fig. 4 shows that internalizing the emission costs of CO_2 results in lower total network internal costs. The reason is that due to the extra costs for CO_2 emissions, some of the flows are transported by rail or IWW at lower CO_2 emission costs despite their extra handling/transhipment costs. The monotonically decreasing curve of the total network internal costs indicates that the costs saved by using rail or IWW instead of road transport can compensate the extra transhipment costs. However, the increasing curve of the total network costs shows that the gain from the above-mentioned modal shift cannot compensate for the internalized costs of CO_2 emissions. The increase in the total network charge of CO_2 emissions is dominant in the change of the total network costs.

Fig. 5 shows the impact of CO_2 pricing on the total network modal shift. The share of road transport decreases from 60% to 20%, while CO_2 emissions are charged from 0 to 1000 €/tCO_2. These percentages are measured in tkm transported in the

Table 1
Description of the analyses setup.

Analysis	Design measures	No. of scenarios	Parameters and their range
1	CO_2 pricing policy	101	• CO_2 price (0–1000 with an increment of 10 €/tCO_2)
2	Terminal network configuration	30,000	• Status of terminals (42 terminals open or closed)
3	Hub-service-networks	2	• Status of hub-service-network (in operation or not)
4	CO_2 pricing and terminal network configuration	30,000	• CO_2 price (0–1000 with an increment of 10 €/tCO_2) • Status of terminals (42 terminals open or closed)
5	CO_2 pricing and hub-service-networks	202	• CO_2 price (0–1000 with an increment of 10 €/tCO_2) • Status of hub-service-network (open or closed)
6	All three measures: CO_2 pricing, terminal network configuration, and hub-service networks	50,000	• CO_2 price (0–1000 with an increment of 10 €/tCO_2) • Status of terminals (42 terminals open or closed) • Status of hub-service-network (open or closed)

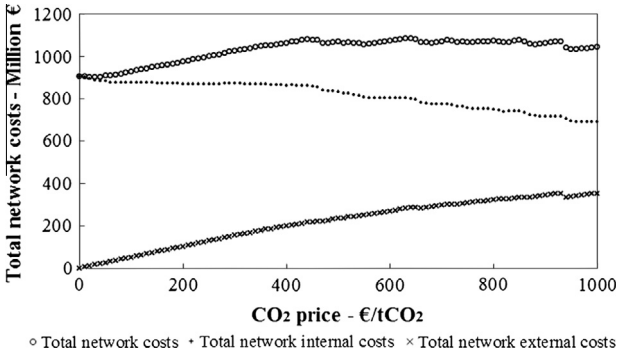


Fig. 4. Relationship between the total network costs and CO_2 prices in the given example.

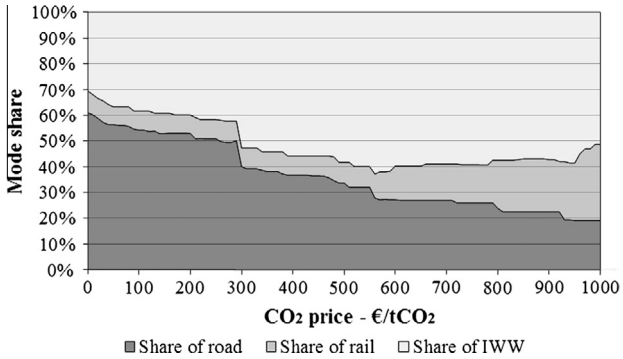


Fig. 5. Share of road, rail, and IWW in the Netherlands measured in tkm in the given example.

Netherlands. The share of road is 40% if measured in tons loaded or/and unloaded in the Netherlands in the base scenario. As illustrated in the Figure, the flows shift first from road to IWW, and then from both road and IWW to rail. This indicates that the IWW transport can receive flows only to a point, after which rail transport takes over.

Although the analysis results cannot explain the exact causation for this pattern, two factors may play a role in such a modal shift pattern. First, the model assumes that most of the locomotives are powered by electricity, and the emissions already counted in the energy sector are not relevant to transport emissions. Therefore, no extra CO_2 emission costs are charged for rail transport. This becomes an overwhelming advantage as compared to road and IWW transport when CO_2 is charged at a high price (around 600 €/tCO_2 in this case). Second, the accessibility of IWW might also limit its competitiveness in contrast to road or rail transport.

3.4.2. Analysis 2: The terminal network configuration

The contribution of reconfiguring the terminal network is exclusively considered under conditions of unavailability of the new hub-service-networks. In each scenario, 42 terminals are assumed to be prospective candidates for container transhipment.

Fig. 6 shows the performances of a large number of configurations of the terminal network. Each dot represents the result of one scenario composed of the combination of a specific terminal network configuration. The dots spread broadly in both dimensions within a relatively large range of the total network costs and the CO_2 emissions, indicating that different terminal configurations result in very different network performances. Many scenarios show better performances than the base scenario, which reveals substantial potential for improvement.

3.4.3. Analysis 3: The hub-service-networks

Fig. 7 shows an example of the cost comparison while moving one container between a given OD pair far away for about 120 km by road/truck, intermodal shuttle barge, and intermodal hub-based-service. As can be seen, the results show the disadvantage of hub-based-services due to increased transhipment costs and lower fleet utilization.

Assuming that hub-based-services are available in the base scenario with a service frequency matching the actual shuttle barge service frequency in 2006, the results show that none of the hub-based-services can feasibly compete with road or

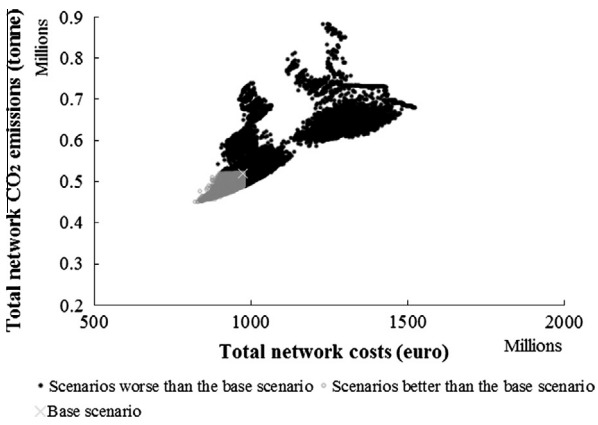


Fig. 6. Relationship between the total network costs and CO_2 emissions. (The results are obtained by using GA; the dots clustering and expanding towards the origin of the quadrant illustrate the clustering and evolving of solutions.)

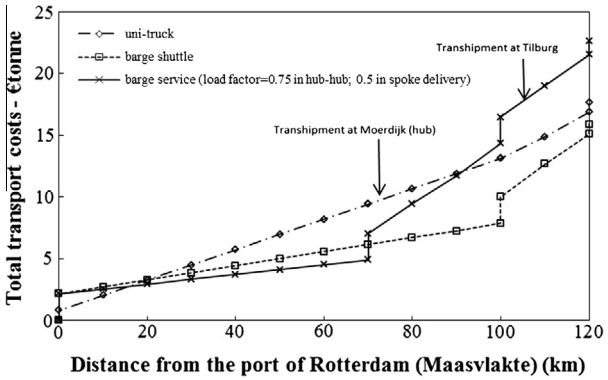


Fig. 7. Relationship between the average total transport costs of a container and distance: cast of the Rotterdam to Tilburg route.

shuttle barge transport services due to the extra transhipment costs and low load factor. The load factor on the links between the inland terminals and the hubs resulting from the model are between 0 and 0.75, whereas the actual load factors of shuttle barge transport services between the inland terminals and the sea terminals vary between 0.6 and 0.9.

3.4.4. Analysis 4: The CO_2 pricing and terminal network configuration

The network performances of 30,000 scenarios are shown in Fig. 8. The series lining the top part of the Figure shows the rather unrealistic scenarios where since no terminal is available in the network, all containers are transported by road. In these cases, the CO_2 price has no impact on the choice of mode or on the total CO_2 emissions. The other series of dots, in the middle of the Figure, shows the scenarios where the terminal network configurations remained the same as in the base scenario. The trend line, for the optimal scenario of each CO_2 price, illustrates the frontier of the minimum total network CO_2 emissions that are achieved by pricing CO_2 at certain rates. The frontier declines as the CO_2 price increases. Comparing these two series and the trend line, the terminal network configuration in the base scenario with intermodal transport has much lower emissions as compared to the scenario where no intermodal transport is available. Optimal terminal network configurations can lead to even lower total network CO_2 emissions. The slope of the line illustrating the scenarios with only CO_2 pricing and the slope of the frontier suggests the effectiveness of each design measure in terms of reducing the total network CO_2 emissions. Consequently, it becomes obvious that combining the two could be more effective than a single measure of CO_2 pricing in terms of total network emissions, especially when CO_2 emissions are charged at a lower rate. For example, when CO_2 is charged at 200 €/tCO_2, the minimum total network emissions under an optimized terminal network configuration are about 60% of the emissions under the base terminal network configuration.

3.4.5. Analysis 5: The CO_2 pricing and hub-service-networks

Fig. 9 shows that hub-based-services start to gain their share of flows when the CO_2 price is around 100 €/tCO_2. The total flows transported by hub-based-services show a tendency to increase with increasing of the CO_2 prices from zero to around 250 €/tCO_2, but also to decrease after this (turning) point. This indicates that with a price lower than 1000-€/tCO_2 and under conditions of constant demand, the share of hub-based-services would be limited within a certain range. Collaborative hub-based-services in IWW would not be competitive to road or shuttle barge transport services in the base scenario in

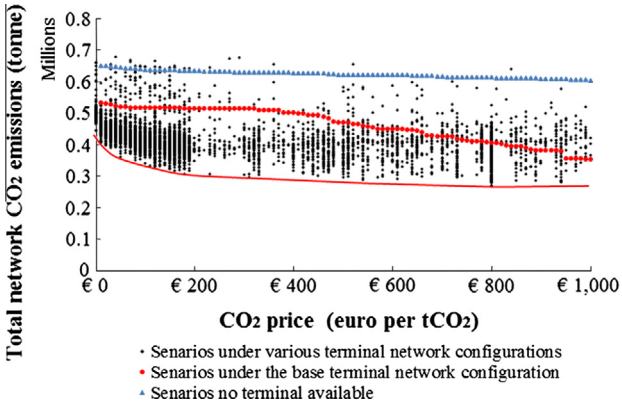


Fig. 8. Relationship between the total network CO_2 emissions and prices of CO_2 emissions.

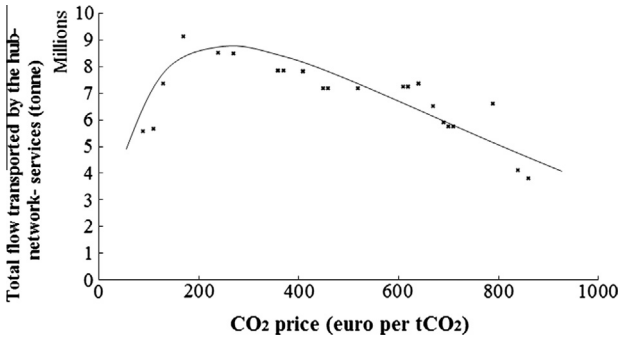


Fig. 9. Relationship between the total flow transported by hub-based-services and different CO_2 prices.

terms of costs. However, they could become feasible if CO_2 pricing was applied at the same time. Higher CO_2 prices could make inland waterway transport more advantageous as compared to road/truck transport, mainly due to its lower CO_2 emissions per tkm (ton-kilometer). The increase in demand for inland waterway transport may lead to feasible hub-based-services. Furthermore, due to the advantage in economies of scale and/or economies of density, hub-based-services may benefit more from an increase in transport demand than shuttle barge transport services.

3.4.6. Analysis 6: The CO_2 pricing, terminal network configuration, and hub-service networks

Fig. 10 shows the total costs and CO_2 emissions for a large number of different scenarios. The series lined up on the top part of this Figure shows the scenarios where no terminal is available in the network. Curve 1 illustrates the frontier of the optimal scenarios, consisting of the optimal terminal network configurations for CO_2 prices ranging from 0 to 1000 €/tCO_2. Curve 2 shows the frontier of the scenarios where all policies are available. Comparing these two frontiers indicates that the combined effects of the three design measures would be more effective in reducing CO_2 emissions than otherwise. The preferred terminal network configuration depends on the CO_2 price.

Moreover, if the terminal network achieved an optimal configuration, transport operators would collaboratively provide appropriate hub-based-services to reduce CO_2 emissions through CO_2 pricing. However, these may not result in an increase in the total network costs. The frontier provides more options in optimization of the terminal network in terms of the target performance.

3.5. Summary and discussion

The above-mentioned results indicate that an increase in the CO_2 emission price leads not only to a reduction in the total network CO_2 emissions, but also to an increase in the total network costs. The benefits gained from using intermodal transport do not compensate for the extra costs of CO_2 emission charges. Therefore, the CO_2 pricing policy may contribute to a reduction of the CO_2 emissions especially when CO_2 is charged at a high rate. However, the extra costs would result in a significant increase in the operational costs directly imposed on transport operators.

Optimal terminal network configurations may result in a network with lower total network costs and lower CO_2 emissions. The assumed hub-based-services cannot compete with road transport or shuttle barge transport services in

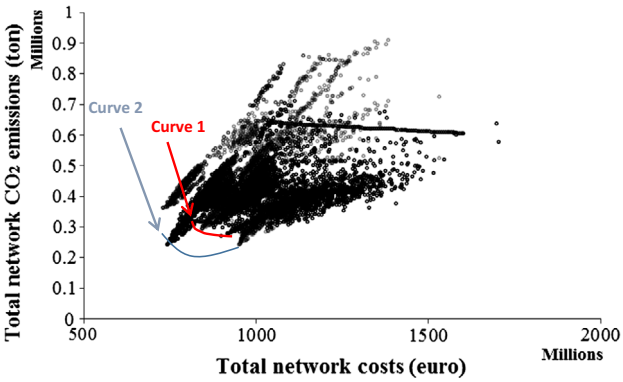


Fig. 10. Relationship between the total network costs and the total network CO_2 emissions for different terminal network configurations and potential hub-based-services. (The results are obtained by using GA; the dots clustering and expanding towards the origin of the quadrant illustrate the clustering and evolving of solutions.)

the base scenario due to the extra transhipment costs and low load factor. They could be more competitive if the demand for inland waterway container transport was much higher or if CO_2 emissions were charged at a sufficiently high price.

A combination of CO_2 pricing and terminal network configuration is more effective than only CO_2 pricing, especially when CO_2 emissions are charged at lower rates. The evaluation analyses show that the combined effect of the three measures is even more effective in reducing CO_2 emissions. This results in a frontier of the minimal total network costs and the total network CO_2 emissions instead of one single optimal solution. It provides more options for optimizing the terminal network in terms of the target network performances. In addition, if the terminal network could achieve an optimal configuration and assuming transport operators would collaboratively provide appropriate hub-based-services to reduce CO_2 emissions through CO_2 pricing, the total network costs may not increase.

4. Conclusions

This paper presents an innovative freight transport network design model, thus positioning itself beyond the existing models in terms of simultaneously incorporating large-scale, multimodal, multi-commodity, multi-actor, and service networks. This model facilitates combinations of design of the freight transport infrastructure networks with respect to pricing policies, terminal network configuration, and collaborative hub-based services, thereby enabling integrated infrastructure, service, and policy design. Design problems have been solved by bi-level optimization, where the upper-level searches for the optimal combinations of policy measures, while the lower-level performs multi-commodity flow assignment over a large-scale multimodal network with two pre-processing procedures. The latter deal with cost efficiency at terminals and collaborative hub-based-services. Optimal fleet, frequency, and choice of terminal used for each service in the service networks are determined based on the flows transported.

The model has been calibrated and validated for a case study of hinterland container transport in the Netherlands using real-life supply and demand data. The results provide new insights into the interrelationships of the infrastructure network, service network, and regulatory policies, as well as the interaction among the different actors.

The model could generically be applicable to freight transport infrastructure network design in terms of architecture, methods, and algorithms. The inclusion of externalities is not limited to CO_2 and could also include other greenhouse gasses (GHG) such as SO_x, NO_x, and social impacts such as noise and traffic incidents/accidents. Users of the model could set the design objectives in accordance with the model application. Evaluation of the network performance could be carried out at the link, terminal, regional, and/or network level, per mode, per commodity type, and/or a combination of the former.

A potential direction for further research could be to incorporate dynamics in both service demand and supply. Although transport demand is a given in this model, pricing policies could have an impact on transport demand. In addition to the existing features of the model, a further development of this model could include capturing the relation between transport demand and transport costs.

Appendix A. Virtual network representation

The virtual network can be described as $Z = (N, L)$ where N and L denote the identifications (IDs) of the nodes of the network and the links of the network, respectively. Each link is associated with its begin-and end-node. Each node is identified by its geographical location, i.e., geographic coordinates. The nodes representing the ends of the real geographic links (i.e., a segment of road, rail, or IWW) are denoted as n_i^m $(i \in N, m \in M)$. m denotes the mode of the geographic link to which the node belongs, M = {road, rail, IWW}. The nodes representing the terminals are denoted as n_i^t $(i \in N, t \in T)$. A terminal is tagged by a terminal ID t. Set T contains the IDs of all terminals. The accessible modes of the terminal are attached to $t \in T$. Besides the fore-/end-nodes of the real geographic links and the terminals, a centroid n_i^o $(i \in N, o \in O)$ is identified by the geographic coordinates of the geographic geometric centroid of the region. The centroids are virtual nodes in the network. They do not represent any infrastructure but each centroid represents the location of the transport demand of a region, and thus each centroid is associated with the ID of a region, $o \in O$. O consists of the set of the IDs of all regions.

The virtual network is designed by connecting the nodes according to the certain rules. Using the categorization of the nodes, links can be categorized automatically based on the node types of their fore-nodes and end-nodes. In this module, five sets of links are generated, $L = X \cup H \cup G \cup A \cup S$. The link l_x^m $(x \in X \subset L, m \in M)$ represents the real geographic link x of mode m. The transhipment links $l_h^{t,m(t)}$ $(h \in H \subset L, t \in T, m \in M)$ are generated by connecting the terminal node n^t to a number of the closest nodes of mode m. $m(t)$ represents the modes which are accessible at the terminal t. The links that describe the pre-/end-haulage between terminal t and centroid o are denoted as $l_g^{o,t}$ $(g \in G \subset L, t \in T, o \in O)$. They are generated by connecting the centroid n^o to the terminals located within a certain geographic radius of centroid o. The radius is region-specific and is determined by the maximum pre-end/haulage of the shipments starting or ending in this region. The centroids are also connected to the terminals within a predefined radius by virtual links (centroid-terminal connector). The access/egress links l_a^o $(a \in A \subset L, o \in O)$ describe the access/egress between the road network and region o. The length of an access/egress link indicates the average distance of accessing from any place of the region to the road network and egressing from an appropriate segment of the road network to a destination in the region. The fifth type of links is service link, which are also virtual links, particularly defined for the service network design. Each service link is denoted as

$l_s^{t_i,t_j}$ $(s \in S \subset L, t \in T)$ represents a leg of one or more service(s) between terminal t_i and t_j. A service link may be parallel to one geographic link or several adjacent geographic links of the same mode.

Appendix B. Examples of the structure of service network

There structures of service networks considered are hub and spoke, hub and pickup-delivery, and hub and circular pickup-delivery service networks. It should be mentioned that more structures exist in practice (Woxenius, 2007). Nevertheless, the above-mentioned three are chosen to be elaborated because they can well reflect the geographic conditions of the presented case – Dutch inland waterways.

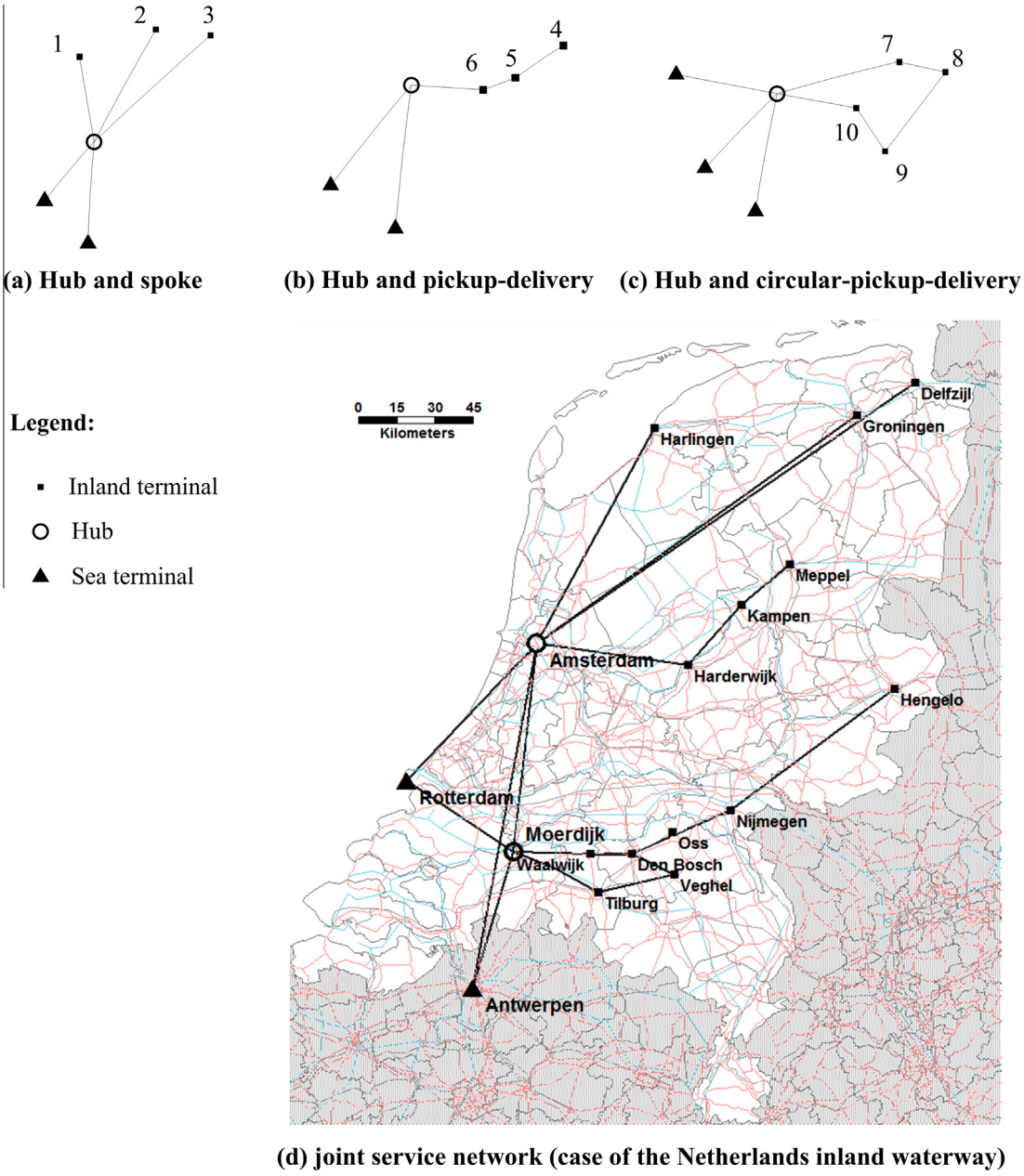


Fig. 11. A simplified scheme of the service network structures, and the hub-based service networks for inland waterway container transport (case of the Netherlands).

B.1. Simple hub and spoke service network (Fig. 11a)

Each inland terminal is connected to the hub via different waterways, and only the hub is directly connected to the sea terminals.

In the case study, the containers are transported from the inland terminals 1, 2, or 3, to the hub (Amsterdam) by small barges, then consolidated and transhipped by larger barges at the hub (Amsterdam), and further transported to the sea terminals.

B.2. Hub and pickup-delivery service network (Fig. 11b)

The inland terminals located along the same waterway form a corridor. The hub is located at one end of the corridor, and as such represents the only terminal providing direct connections to the sea terminals.

In the case study, the transport services are provided by small barges in the sequential order $4 \rightarrow 5 \rightarrow 6$ to the hub (Amsterdam) and in the sequential order $6 \rightarrow 5 \rightarrow 4$ from the hub (Amsterdam). Containers are loaded at each inland terminal, consolidated at the hub (Amsterdam), and further transported to the sea terminals by larger barges, and vice versa.

B.3. Hub and circular-pickup-delivery service network (Fig. 11c)

This structure is similar to the structure (b), but the main difference is that the inland terminals are located along a circular waterway.

In the case study, the transport services are provided in the sequential order of $7 \rightarrow 8 \rightarrow 9 \rightarrow 10$, both in the clockwise and counter-clockwise direction by small barges. Containers are loaded and unloaded at each terminal, consolidated to the larger barges at the hub (Moerdijk), and further transported to the sea terminals.

A simplified scheme of the integrated hub service networks for the container transport by inland waterway in the Netherlands is shown in Fig. 11d.

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